

Rongzhi Dong

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Summary

Skilled programmer with experience in **Python/MATLAB/C**. Experienced researcher using **Numpy, Pandas**, and **Scikit-learn** for data analysis; **RFE, XGBoost**, and **LightGBM** for feature selection; and **CNN, GAN, GNN, ResNet, LSTM, Transformer**, and **Diffusion** models for algorithm design. Motivated learner with a strong command of **Deep Learning** and **Machine Learning** principles. Collaborative partner with an analytical mind for problem-solving.

Skills

Deep learning • Machine Learning • AI for Material

Languages	📖 Python(proficient) · MATLAB(experienced) · C(experienced) · Shell · Git
Programming	📖 PyTorch · Scikit-learn · TensorFlow · Keras
Simulation	📖 ANSYS · ABAQUS · SolidWorks · MATLAB
Communication	📖 Writing · Presentation
Courses	📖 Data Mining and Warehousing · Computer Vision · Pattern Recognition and Machine Learning · Advanced Artificial Intelligence · Analysis of Algorithms

Education

2021 – now	📖 University of South Carolina, Ph.D. student in Computer Science.	Columbia, SC, United States
2018 – 2021	📖 Guizhou University, Master in Mechanical Engineering	Guiyang, Guizhou, China
2013 – 2017	📖 China Agricultural University, Bachelor in Engineering Mechanics.	Beijing, China

Work experience

2024.5 – 2024.8	📖 Lawrence Livermore National Laboratory, Summer Research Intern.	Livermore, CA, United States
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Projects

Diffusion Language Model based Material Generation

- 📖 Developed a diffusion-based pipeline for generating novel materials by amalgamating **diffusion language model**-based composition generators, a template-based crystal structure predictor, and a **graph neural network** potential-based structure relaxation algorithm. This model outperforms the GAN-based generator and demonstrates competitive performance with transformer-based generators.
- 📖 Authored research paper accepted by The Journal of Physical Chemistry A.

Projects (continued)

Transformer-based 2D Material Generation

- Developed a 2D material generation pipeline by integrating a **transformer**-based 2D material composition generator, two template-based crystal structure predictors, and two **graph neural network** potential-based structure relaxation algorithms. And discovered two thermodynamically, mechanically, and dynamically stable new 2D structures.
- Authored research paper accepted by Advanced Intelligent Systems.

Predicting XRD spectrum from Material Composition

- Developed a **CNN** model to predict XRD spectra from material compositions. This model can be used to conduct high-throughput screening of the composition design space for structures with specific structural features or symmetry. Based on the predicted XRD spectra, the material's properties can be estimated.
- Authored research paper published in ACS Applied Materials & Interfaces.

Research Publications

- R. Dong**, N. Fu, E. M. Siriwardane, and J. Hu, "Generative design of inorganic compounds using deep diffusion language models," *The Journal of Physical Chemistry A*, vol. 128, no. 29, pp. 5980–5989, 2024.
- J. Hu, D. Liu, N. Fu, and **R. Dong**, "Realistic material property prediction using domain adaptation based machine learning," *Digital Discovery*, vol. 3, no. 2, pp. 300–312, 2024.
- R. Dong**, Y. Song, E. M. Siriwardane, and J. Hu, "Discovery of 2d materials using transformer network-based generative design," *Advanced Intelligent Systems*, vol. 5, no. 12, p. 2300141, 2023.
- R. Dong**, Y. Zhao, Y. Song, N. Fu, S. S. Omee, S. Dey, Q. Li, L. Wei, and J. Hu, "Deepxrd, a deep learning model for predicting xrd spectrum from material composition," *ACS Applied Materials & Interfaces*, vol. 14, no. 35, pp. 40102–40115, 2022.
- D. Varivoda, **R. Dong**, S. S. Omee, and J. Hu, "Materials property prediction with uncertainty quantification: A benchmark study," *Applied Physics Reviews*, vol. 10, no. 2, 2023.
- Q. Li, **R. Dong**, N. Fu, S. S. Omee, L. Wei, and J. Hu, "Global mapping of structures and properties of crystal materials," *Journal of Chemical Information and Modeling*, vol. 63, no. 12, pp. 3814–3826, 2023.
- Y. Song, **R. Dong**, L. Wei, Q. Li, and J. Hu, "Alphacrystal-ii: Distance matrix based crystal structure prediction using deep learning," *arXiv preprint arXiv:2404.04810*, 2024.
- S. S. Omee, N. Fu, **R. Dong**, M. Hu, and J. Hu, "Structure-based out-of-distribution (ood) materials property prediction: A benchmark study," *npj Computational Materials*, vol. 10, no. 1, p. 144, 2024.
- L. Wei, S. S. Omee, **R. Dong**, N. Fu, Y. Song, E. Siriwardane, M. Xu, C. Wolverton, and J. Hu, "Cspbent: A benchmark and critical evaluation of crystal structure prediction," *arXiv preprint arXiv:2407.00733*, 2024.
- X. Zhang, S. Li, L. Yu, **R. Dong**, Q. Song, and Z. Wang, "Thermal conductivity prediction with fully connected neural network," in *AIIPCC 2022: The Third International Conference on Artificial Intelligence, Information Processing and Cloud Computing*, VDE, 2022, pp. 1–5.
- L. Wei, Q. Li, Y. Song, S. Stefanov, **R. Dong**, N. Fu, E. M. Siriwardane, F. Chen, and J. Hu, "Crystal composition transformer: Self-learning neural language model for generative and tinkering design of materials," *Advanced Science*, vol. 11, no. 36, p. 2304305, 2024.

- 12 J. Hu, Y. Zhao, Q. Li, Y. Song, **R. Dong**, W. Yang, and E. M. Siriwardane, "Deep learning-based prediction of contact maps and crystal structures of inorganic materials," *ACS omega*, vol. 8, no. 29, pp. 26 170–26 179, 2023.
- 13 S. S. Omee, S.-Y. Louis, N. Fu, L. Wei, S. Dey, **R. Dong**, Q. Li, and J. Hu, "Scalable deeper graph neural networks for high-performance materials property prediction," *Patterns*, vol. 3, no. 5, 2022.
- 14 L. Wei, N. Fu, E. M. Siriwardane, W. Yang, S. S. Omee, **R. Dong**, R. Xin, and J. Hu, "Tcsp: A template-based crystal structure prediction algorithm for materials discovery," *Inorganic Chemistry*, vol. 61, no. 22, pp. 8431–8439, 2022.
- 15 N. Fu, L. Wei, Y. Song, Q. Li, R. Xin, S. S. Omee, **R. Dong**, E. M. D. Siriwardane, and J. Hu, "Material transformers: Deep learning language models for generative materials design," *Machine Learning: Science and Technology*, vol. 4, no. 1, p. 015 001, 2023.
- 16 J. Hu, Y. Song, S. S. Omee, L. Wei, **R. Dong**, and S. Gianey, "Evolutionary machine learning in science and engineering," in *Handbook of Evolutionary Machine Learning*, Springer Nature Singapore Singapore, 2023, pp. 535–561.